

HOMEWORK 11

6500-324: DATA MANAGEMENT FOR INFORMATION SYSTEMS (Fall 2022)

©2022 by ecourse.org

Part 1. Multiple Choice or True/False

1. (1 point) Find the department that has a largest average salary
A) select deptno from emp group by deptno having avg(sal) >= all (select deptno, avg(sal) from emp group by deptno);
B) select deptno from emp group by deptno having avg(sal) >= (select max(avg(sal)) from emp group by deptno);
C) select dname from emp group by deptno having avg(sal) >= all (select max(avg(sal)) from emp group by deptno);
D) select deptno from emp group by deptno having avg(sal) >= all (select avg(sal) from emp group by deptno);
2. (1 point) Soundex() function is used to search for records based on the sound of
A) entire cell value
B) individual words in a cell
C) entire record
D) column name
3. (1 point) To select a random number of records in Oracle database, you will use which of the following to randomize the order first:
A) dbms_random.value
B) Random() function
C) Rank() function
D) Rownum
4. (1 point) When running a select statement returns multiple records, how would you receive the data?
A) declare multiple variables to receive the records
B) declare a cursor to receive the records
C) handle many_data_found error in exception section
D) declare a recordset to receive the data
5. (1 point) (select all that apply) In a hierarchical recursive query, with clause is used to build a subquery. What is that subquery different from subqueries used in other clauses such as where or from clause?
A) Subquery in with clause uses union while others do not
B) subquery in with clause uses union all while others do not
C) Subquery in with clause can be self called by the subquery while others cannot
D) There is no difference
6. (1 point) Which of the following is not a data model for NoSQL databases?

- A) key-value pairs
 - B) relational model
 - C) network model
 - D) graphic model
 - E) nested document-based model
7. (1 point) (select all that apply) What are the features of NoSQL databases?
- A) schema-less
 - B) collections are linked by references
 - C) optimized for for a data model that aims at high performance
 - D) store documents instead of records
 - E) balanced approach to availability and consistency
8. (1 point) What is CAP Theorem?
- A) No distributed systems can be scalable as well as consistent
 - B) MongoDB cannot be scalable
 - C) A distributed system can meet only two of the following: consistency, availability, and errors tolerance
 - D) A relational database cannot meet the requirements of big data
9. (1 point) What is isolation property for a distributed system?
- A) All transactions must be done sequentially one by one
 - B) All transactions should lead to the same result whether they are performed concurrently or sequentially
 - C) All transactions can be performed concurrently
 - D) All transactions must all succeed or fail as a whole
10. (1 point) What is the best characterization of JSON?
- A) data format for storing and communication object data by JavaScript
 - B) JavaScript programming language
 - C) data model for MongoDB
 - D) key-value store